Biogeochemical Resilience and Micro-Environmental Homeostasis in Scleractinian Cora

D. H. Johansen, M. E. Lindqvist, S. R. Patel (2024)

Abstract

Recent in-situ mesocosm evaluations demonstrate that energetic surplus modulates physiological responses.

Elevated atmospheric CO2 does not decrease calcification rates in marine calcifiers.

Higher sea surface temperatures are significantly associated with accelerated coral bleaching events.

Discussion & Photo-Physiological Acclimation

Photophysiological performance was sustained through active proton pumping across the sub-calicoblastic space. Thermal acclimation at 28 degrees Celsius increased photosynthetic efficiency in Symbiodiniaceae. Our findings highlight that heterotrophic nutrition confers substantial resilience to acidification.